Communication Methods in Digital Systems Lab 5

AXI Bus - Extended

Introduction

This laboratory is intended to illustrate the advanced communication properties of the AXI bus system. There will be analyzed designs of the interface system that enables capturing image data from a camera and placing it into memory using the AXI master interface. The second case shows the opposite path that acquires the image data from the memory and transfers it to HDMI controller module.

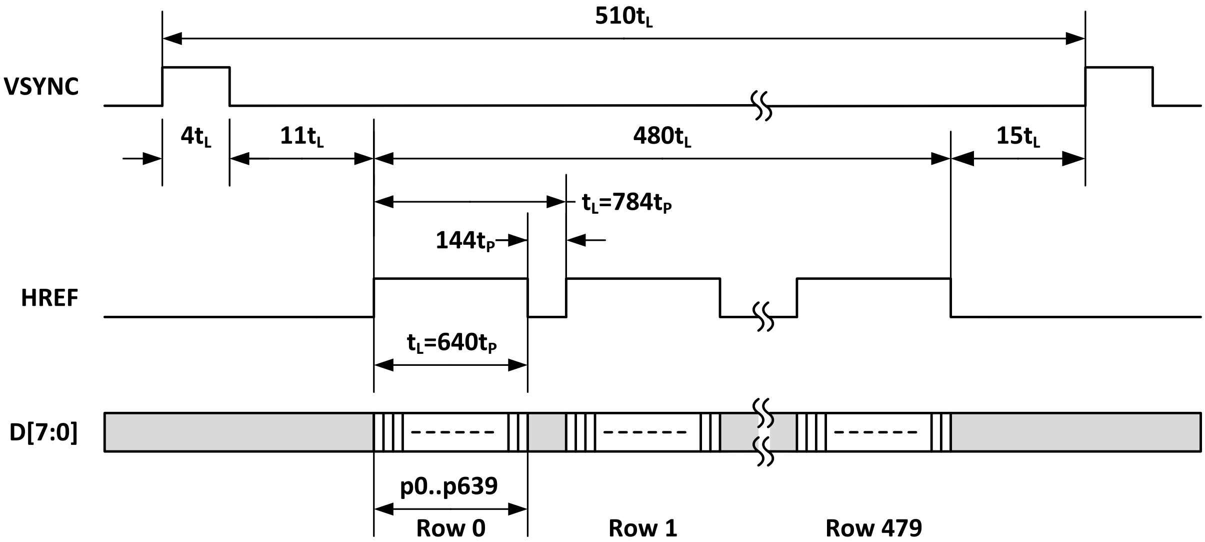

Fig. 1. The OV7660 image data stream waveform

The OV7660 is a digital camera chip with a parallel interface. The data stream from the sensor is delivered according to the waveforms shown in Fig. 1. There are vertical and horizontal synchronization signals. The data format resembles composite analog video signal (decomposed into elementary signals). The OV7660 in the design is replaced by the functional module that produces respective signals according to delivered clock signal.

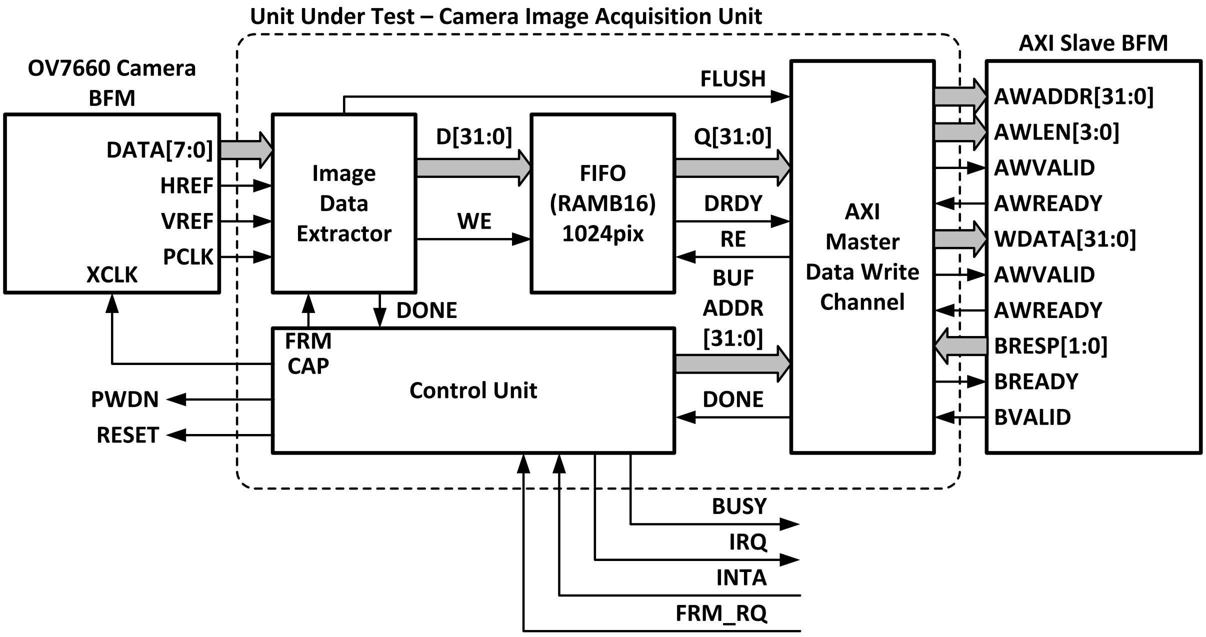

Fig. 2. The OV7660 camera interface to AXI bus controller connection (synthesizable) and AXI device functional model

The data from the camera requires passing to the memory using the AXI interface. The exemplary protocol translator between camera interface and AXI bus system is shown in Fig. 2. The goal of the laboratory is to design the protocol translator. In order to assure performance efficient implementation the data bursts are used. The implementation utilize maximal burst size of 16 words (AXI-3). The functional model

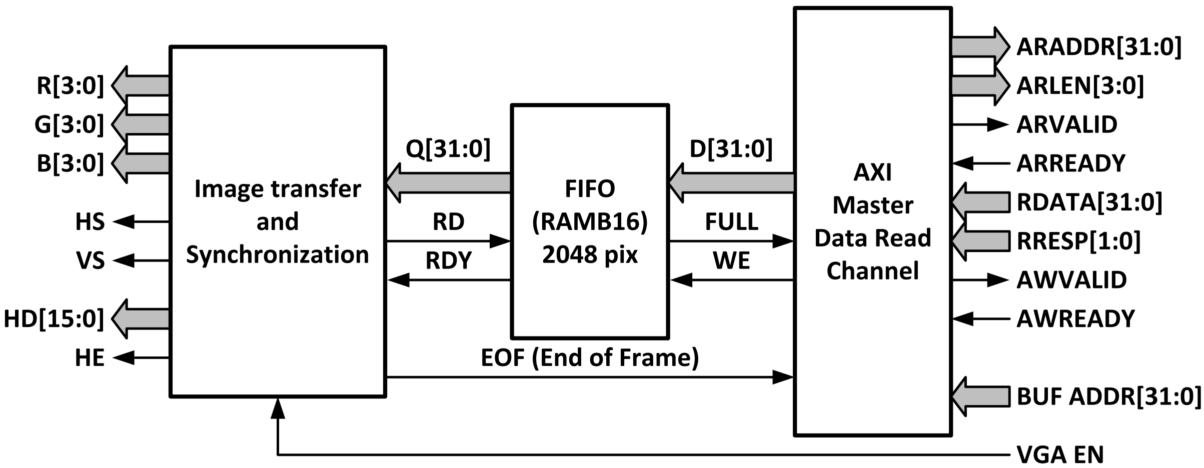

Fig. 3. The VGA interface to AXI bus controller connection (synthesizable).

The second example is driving the HDMI display controller with data from the memory using the AXI master controller. The developed interface bridge is supposed to collect the data from memory and transfer it to HDMI display controller. Implemented internal buffer must be always filled with data to assure continues and smoth image displaying. The display process cannot be started sooner than collecting enough image data items.

AXI data transfer

1. Design an interface to the OV7660 camera that transfers data to the memory. Use the OV76_BHV model to simulate the camera interface behavior. The Initial part of the AXI master controller along with the interface to the camera is placed in OV76_TO_AXIM. The module is dedicated to capturing a single image frame from the camera. Upon asserting the FRM_RQ signal a new frame should be collected. To speed up the data transfer a burst transmission is used. The data from the camera must be collected in the temporary buffer. The data is transmitted using the maximal burst size of 16 words (32-bit words). Implement a FIFO that allows buffer data arriving from the camera. Perform data transfer only if the FIFO contains enough data items to be transferred to memory. The AXI device is simulated by the functional model that will respond to requests from the designed controller. Try to write the controller in a manner that assures its hardware implementation.
2. Implement a controller that reads data from the memory and transfers it to the HDMI controller. The controller should be implemented inside the AXIM_TO_HDMI module. There has been prepared testbench (AXIM_TO_HDMI_tb) to simplify the development and verification process. For your disposal is the device AXI interface that responds with data for image controller. Your goal is to propose the implementation of the AXI controller only.

All the examples are prepared for simulation with use of Icarus Verilog and GTKWave viewer. There are prepared respective batch files that compile designs, run simulation and finally displays the results in the waveform viewer.